

MOHAMMED TAHMID HOSSAIN

AI Data Quality Specialist • LLM Evaluator • Machine Learning Researcher
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PROFESSIONAL SUMMARY

Data Quality Specialist and AI Researcher with 2+ years of experience in multimodal data annotation, LLM evaluation, and AI training workflows. Published author at IEEE international conferences with applied research in machine learning, environmental analytics, and healthcare cybersecurity. Proven expertise in rubric-based QA, large-scale dataset validation, and mentoring AI trainers. Proficient in Python, cloud platforms (AWS, GCP), and data tools.

WORK EXPERIENCE

Research Assistant (Remote) | Center for Computational & Data Sciences

Jan 2024 – Present
Dhaka, Bangladesh

- Conducted applied research on IoMT cybersecurity, developing machine learning classifiers with information-gain feature selection to detect and categorize cyberattacks on Internet of Medical Things devices; published at IEEE ASET 2024.
- Led computer vision research evaluating YOLOv8, YOLOv10, and YOLOv12 object detection models on the Poribohon-BD vehicle dataset under resolution scaling, JPEG compression, and Real-ESRGAN restoration; published at IEEE SCReD 2025.
- Collaborated with multidisciplinary research teams, contributing to data preprocessing, model training, experimental analysis, and academic manuscript preparation.

QA – Senior AI Data Trainer | Invisible Technologies Inc.

Sep 2024 – Sep 2025
San Francisco, CA

- Cleaned, optimized, and quality-checked 10,000+ multimodal data samples (text, image, and instruction) to improve LLM training performance.
- Performed rubric-based QA evaluation of LLM outputs for accuracy, relevance, safety, and instruction-following across large-scale pipelines.
- Mentored and onboarded new AI trainers through calibration reviews, structured feedback, and quality guidance, improving team consistency.
- Authored requirement and process documentation using Google Docs to standardize annotation and QA workflows across the team.
- Tracked quality metrics and reviewer feedback in Google Sheets using research-oriented evaluation methods to identify model behavior trends and data quality issues.

AI Data Trainer | Outlier AI

Mar 2024 – Aug 2024
Oakland, CA

- Created and evaluated multilingual (Bengali–English) prompts and responses to improve LLM accuracy, fluency, and cultural alignment.
- Conducted QA reviews of prompt–response pairs, identifying factual, linguistic, and instruction-following issues.
- Logged annotation outcomes and reviewer feedback in Google Sheets to support iterative model improvement.

Teaching Assistant (TA) | Independent University, Bangladesh

Feb 2021 – Aug 2022
Dhaka, Bangladesh

- Mentored 200+ undergraduate students in programming and data structures, supporting coursework in algorithms and software development.

EDUCATION

M.S. Computer Science – Big Data Systems
Arizona State University

Expected May 2027
Tempe, AZ

OTHER EXPERIENCE

Residential Associate | BHRAGS Alliance Inc.

Nov 2025 – Present
New York, NY

- Supported day-to-day operations in a regulated residential environment while maintaining compliance and documentation standards.

PUBLICATIONS

- [1] Hossain, M.T. et al. *Resolution, Compression, and GAN-Enhanced Evaluation of YOLO Models on the Poribohon-BD Vehicle Dataset*. **IEEE SCORED 2025**, Malaysia.
- [2] Hossain, M.T. et al. *Cyberattacks Classification on Internet of Medical Things Using Information Gain Feature Selection and Machine Learning*. **IEEE ASET 2024**, UAE.
- [3] Hossain, M.T. et al. *Understanding the Dynamics of Dengue in Bangladesh: EDA, Climate Correlation, and Predictive Modeling*. **IEEE TENCON 2023**, Thailand.
- [4] Hossain, M.T. et al. *Impact of COVID-19 Lockdowns on Air Quality in Bangladesh: Analysis and AQI Forecasting with Support Vector Regression*. **IEEE INCET 2023**, India.

SELECTED PROJECTS

Student Performance Monitoring System

- Built a cloud-based web application using Node.js (Express) and SQLite to track and analyze academic performance at Independent University, Bangladesh.
- Enabled faculty to monitor student progress efficiently with real-time data views. — github.com/saditahmid/SPM_Advanced

AQI Exploratory Data Analysis

- Analyzed Bangladesh's Air Quality Index (2014–2023) across 8 major locations using Python to study the environmental impact of COVID-19 lockdowns.
- Visualized environmental trends to support data-driven policy discussions. — saditahmid.github.io

SKILLS

Data & AI: Multimodal annotation, LLM evaluation, rubric-based QA, dataset validation, RLHF, prompt engineering

Research: Exploratory Data Analysis (EDA), feature selection, predictive modeling, statistical analysis

Programming: Python, C++, HTML, CSS, JavaScript (Node.js)

Libraries & Frameworks: NumPy, Pandas, OpenCV, Scikit-learn, Express.js

Cloud & DevOps: AWS, Google Cloud Platform (GCP), Docker, Kubernetes

Tools: Google Sheets, Google Docs, Git, Linux, Jupyter Notebook